

HODGE THEORY

# The $\bar{\partial}$ -Poincaré lemma

One complex dimension

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# The equation we want to solve

Let  $D \subset \mathbb{C}$  be a disk, with coordinate  $z = x + iy$ .

Given a smooth  $(0, 1)$ -form

$$\eta = g(z, \bar{z}) d\bar{z}$$

extending smoothly to a neighbourhood of  $\bar{D}$ , we want

$$\bar{\partial}u = \eta, \quad u \in C^\infty(D).$$

In coordinates this is just

$$\bar{\partial}u = \frac{1}{2}(u_x + iu_y) d\bar{z}.$$

We first solve the equation on a torus up to a constant term.

Hodge theory

# Functions on the square torus

We start with the elliptic curve

$$E = \mathbb{C}/(2\pi\mathbb{Z} + 2\pi i\mathbb{Z}).$$

Think of  $(0, 2\pi] \times (0, 2\pi]$  with opposite edges identified.

A function on  $E$  is a function on  $\mathbb{C}$  satisfying

$$f(x + 2\pi, y) = f(x, y), \quad f(x, y + 2\pi) = f(x, y).$$

There are two real coordinates, so there will be two frequencies.

# The Fourier basis

The functions

$$e_{n,m}(x, y) = e^{i(nx+my)} = e^{inx} e^{imy}, \quad (n, m) \in \mathbb{Z}^2,$$

form an orthonormal basis of  $L^2(E)$  for the measure  $dx dy / (2\pi)^2$ .

Therefore every  $f \in L^2(E)$  has a basis expansion

$$f = \sum_{n,m \in \mathbb{Z}} a_{n,m} e_{n,m}, \quad a_{n,m} = \langle f, e_{n,m} \rangle.$$

The equality means convergence in  $L^2$ . We use the Fourier basis theorem without proving it here.

# Fourier modes of a form

Expand the coefficient of a smooth  $(0, 1)$ -form:

$$\eta = g d\bar{z} = \sum_{n,m \in \mathbb{Z}} \underbrace{a_{n,m} e_{n,m}}_{\eta_{n,m}} d\bar{z}.$$

The summand  $\eta_{n,m}$  is its  $(n, m)$ -Fourier mode.

$$\eta = a_{0,0} d\bar{z} + \sum_{(n,m) \neq (0,0)} \eta_{n,m}.$$

The zero mode has constant coefficients. A nonzero frequency means  $(n, m) \neq (0, 0)$ .

# Differentiation preserves the frequency

Differentiate one basis function:

$$de_{n,m} = e_{n,m}(in\,dx + im\,dy),$$

$$\bar{\partial}e_{n,m} = \frac{in - m}{2} e_{n,m} d\bar{z}.$$

Every coefficient is a constant multiple of the same  $e_{n,m}$ .

Thus  $d$  and  $\bar{\partial}$  preserve the pair  $(n, m)$ . We can solve the equation one frequency at a time.

# Closed forms and exact forms

A form  $\eta$  is  $\bar{\partial}$ -closed if  $\bar{\partial}\eta = 0$ .

It is  $\bar{\partial}$ -exact if  $\eta = \bar{\partial}u$  for some  $u$ .

In complex dimension one, every  $(0, 1)$ -form is closed:

$$\bar{\partial}(g d\bar{z}) = \frac{\partial g}{\partial \bar{z}} \underbrace{d\bar{z} \wedge d\bar{z}}_0 = 0.$$

Our task is to find primitives. Closedness is already automatic in this degree.

# The complex Cartan formula

Put  $X = \partial_x$ , so  $IX = \partial_y$ .

For these constant vector fields,

$$\bar{\partial}\iota_X + \iota_X\bar{\partial} = \frac{1}{2}(\mathcal{L}_X + i\mathcal{L}_{IX}).$$

Here  $\iota_X$  is contraction. On a one-form,  $\iota_X\eta = \eta(X)$ . In particular,

$$\iota_X(g\,d\bar{z}) = g.$$

Here  $\mathcal{L}_X$  and  $\mathcal{L}_{IX}$  simply differentiate the coefficients in the  $x$ - and  $y$ -directions.

# Primitives of nonzero modes

For  $\eta_{n,m} = a_{n,m} e_{n,m} d\bar{z}$ , we have

$$\mathcal{L}_X \eta_{n,m} = in \eta_{n,m}, \quad \mathcal{L}_{lX} \eta_{n,m} = im \eta_{n,m}.$$

Since  $\bar{\partial} \eta_{n,m} = 0$ , Cartan's formula gives

$$\bar{\partial}(\iota_X \eta_{n,m}) = \frac{in - m}{2} \eta_{n,m}.$$

If  $(n, m) \neq (0, 0)$ , then  $in - m \neq 0$ , hence

$$\eta_{n,m} = \bar{\partial} \left( \frac{2a_{n,m}}{in - m} e_{n,m} \right).$$

# Adding the primitives

Define a function on the torus by

$$f = \sum_{(n,m) \neq (0,0)} \frac{2a_{n,m}}{in - m} e_{n,m}.$$

Differentiating term by term gives

$$\bar{\partial}f = \eta - a_{0,0} d\bar{z}.$$

Therefore

$$\eta = c d\bar{z} + \bar{\partial}f, \quad c = a_{0,0}.$$

Fourier fact used here: for smooth  $g$ , this series and all its derivatives converge uniformly, so  $f$  is smooth.

Hodge theory

# The constant part in cohomology

Dolbeault cohomology identifies closed forms that differ by an exact form:

$$H_{\bar{\partial}}^{0,1}(E) = \frac{\{\text{smooth closed } (0,1)\text{-forms}\}}{\{\bar{\partial}f : f \in C^\infty(E)\}}.$$

Since  $\eta = c d\bar{z} + \bar{\partial}f$ ,

$$[\eta] = c[d\bar{z}], \quad H_{\bar{\partial}}^{0,1}(E) \cong \mathbb{C}.$$

$\bar{\partial}f$  always has zero constant part: differentiation preserves frequencies and kills the constant term. Thus  $c d\bar{z}$  is exact only when  $c = 0$ .

# The disk inside an elliptic curve

Choose  $L > 0$  so large that  $\bar{D}$  lies in the interior of a fundamental square for

$$E_L = \mathbb{C}/(L\mathbb{Z} + iL\mathbb{Z}).$$

The quotient map embeds a neighbourhood of  $\bar{D}$  into  $E_L$ .

The same Fourier calculation works for any period  $L$ .

Rescaling the coordinate by  $w = 2\pi z/L$  reduces it to the square torus we just used.

On this neighbourhood,  $z$  and  $\bar{z}$  are well-defined functions.

# Extending the form to the torus

Let  $U$  be a neighbourhood of  $\overline{D}$  inside that square, where  $\eta$  is defined.

Choose a smooth function  $\chi$  that is 1 near  $\overline{D}$  and has support contained in  $U$ .

$$\tilde{\eta} = \begin{cases} \chi\eta & \text{on } U, \\ 0 & \text{outside } U. \end{cases}$$

This is a smooth  $(0, 1)$ -form on  $E_L$ , equal to  $\eta$  on  $D$ .

It is automatically  $\bar{\partial}$ -closed, just like every  $(0, 1)$ -form on a complex curve.

# The primitive on the disk

On the elliptic curve, our calculation gives

$$\tilde{\eta} = c d\bar{z} + \bar{\partial}f.$$

On the disk,  $\bar{z}$  is a function and  $\bar{\partial}\bar{z} = d\bar{z}$ . Restricting to  $D$  therefore gives

$$\eta = \bar{\partial}(c\bar{z} + f|_D).$$

This is the required primitive.

On the whole torus,  $\bar{z}$  is not periodic, so it cannot serve as a global primitive.

# Supplement: the $(1, 1)$ case

A  $(1, 1)$ -form on the disk is  $\eta = g \, dz \wedge d\bar{z}$ .

By the  $(0, 1)$  case, choose  $u$  with

$$\bar{\partial}u = g \, d\bar{z}.$$

Then

$$\bar{\partial}(-u \, dz) = -\bar{\partial}u \wedge dz = g \, dz \wedge d\bar{z} = \eta.$$

These are the two possible types with positive antiholomorphic degree in complex dimension one:  $(0, 1)$  and  $(1, 1)$ .

# Supplement: the constant representatives

The same calculation for each type gives

| Type     | Constant representative | Cohomology on $E$                              |
|----------|-------------------------|------------------------------------------------|
| $(0, 0)$ | $1$                     | $H_{\bar{\partial}}^{0,0}(E) \cong \mathbb{C}$ |
| $(1, 0)$ | $dz$                    | $H_{\bar{\partial}}^{1,0}(E) \cong \mathbb{C}$ |
| $(0, 1)$ | $d\bar{z}$              | $H_{\bar{\partial}}^{0,1}(E) \cong \mathbb{C}$ |
| $(1, 1)$ | $dz \wedge d\bar{z}$    | $H_{\bar{\partial}}^{1,1}(E) \cong \mathbb{C}$ |

Here  $(p, q)$  records the number of  $dz$  and  $d\bar{z}$  factors. Each cohomology class has a unique constant representative.